

09 What's Broken

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10 MIN READ · INTERACTIVE

Scenes do not fail all at once, and the properties that survive longest are the ones a per-frame score rewards. What breaks, in what order, and why nobody can tell you which of these systems is best.

Every chapter so far has ended with something that does not work yet.

This one collects them, because the gap between what these systems can do and what the announcements imply is where most of the confusion in this subject lives, and because knowing what is broken is the difference between reading a paper and being sold one.

Rollouts do not fail all at once

The usual way to report how long a model holds up is an average error over a rollout. That number is close to useless, because the properties of a scene break in an order, and the order is not the one you would guess.

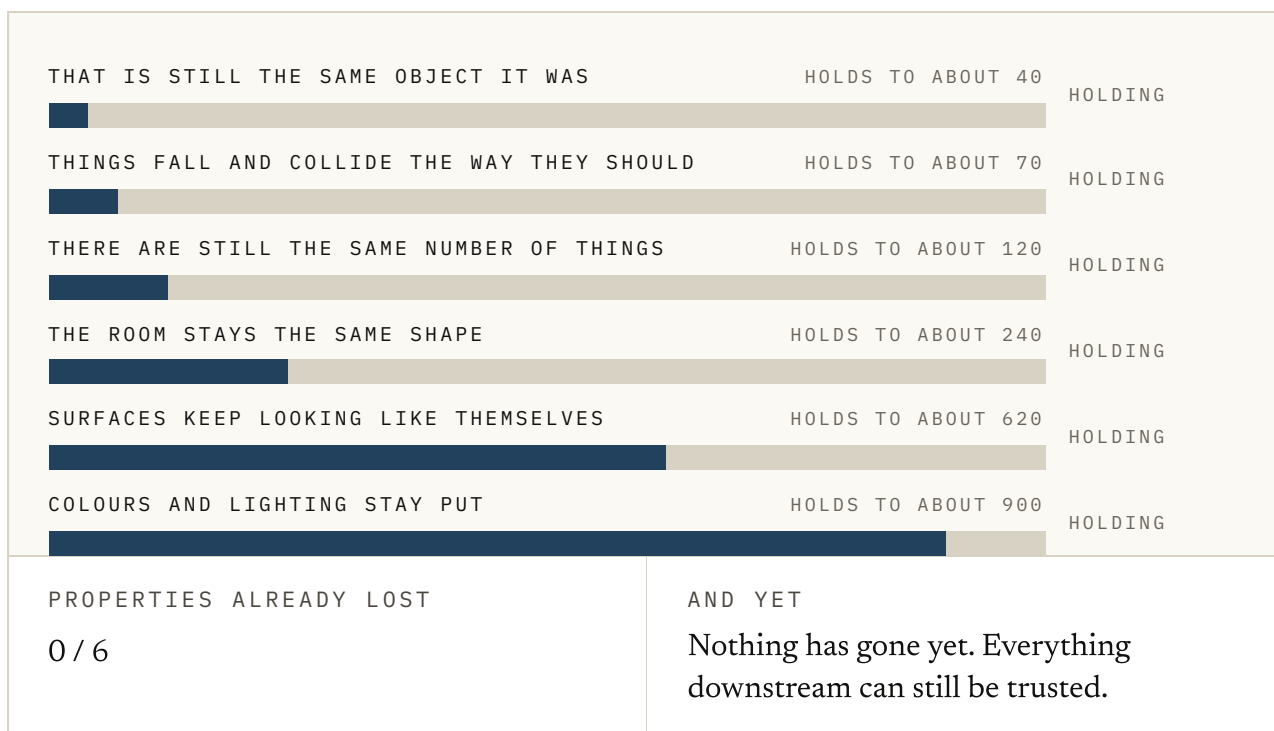


FIG. 9.1 The horizons are illustrative of the ordering rather than measurements of any named system. What is not illustrative is the shape: things go in roughly this sequence, and the ones that survive longest are the ones a per-frame score happens to reward. Drag out to a few hundred steps and read the bottom right.

Colour and lighting are stable almost indefinitely. Texture holds up well. What goes first is whether that is still the same object it was a moment ago, and whether things are still behaving as though they have mass.

Which is precisely backwards from what you want, and it is not a coincidence. A loss computed per frame rewards getting most of the pixels approximately right, and most of the pixels are surfaces and light. Nothing in that loss is watching whether the chair you walked past is the same chair.

So a rollout can be a hundred steps past the point of being useful and still look, in any single frame, completely fine. That is the specific way this subject is easy to fool yourself about.

The three that keep coming back

Things stop being themselves. Object permanence is the one everyone notices, and it is worth being precise about why it is hard. Nothing in these systems is storing an object. Consistency is a property of a well-trained predictor, so it degrades the way predictions degrade, gradually and without warning. There is no error message when the chair becomes a different chair.

What if is not really available. A model can tell you what is likely to happen next. Asking what would have happened if you had done something else is a different question, and the answer you get is true inside the model rather than about the world. Where the model has data the two are close. Where it does not, they are not, and nothing in the answer indicates which case you are in.

This one is easy to under-rate because the model always answers. A system that said "I have never seen anything like that" would be more useful and less impressive, and nobody is rewarded for building it.

Long horizons are still the binding constraint. Every trick in this course is, read uncharitably, a way of not needing the model to be right for very long. Short rollouts. Replanning. Starting from real states. They work. They also amount to an admission that the thing does not hold up, and the horizon over which it does is the number that decides what you can build.

Nobody can tell you which one is best

This is the part that should bother you most, and it gets the least attention.

There is no agreed way to measure whether a generated world is behaving. There are several disagreeing ways, and a system that wins on one routinely loses on another.

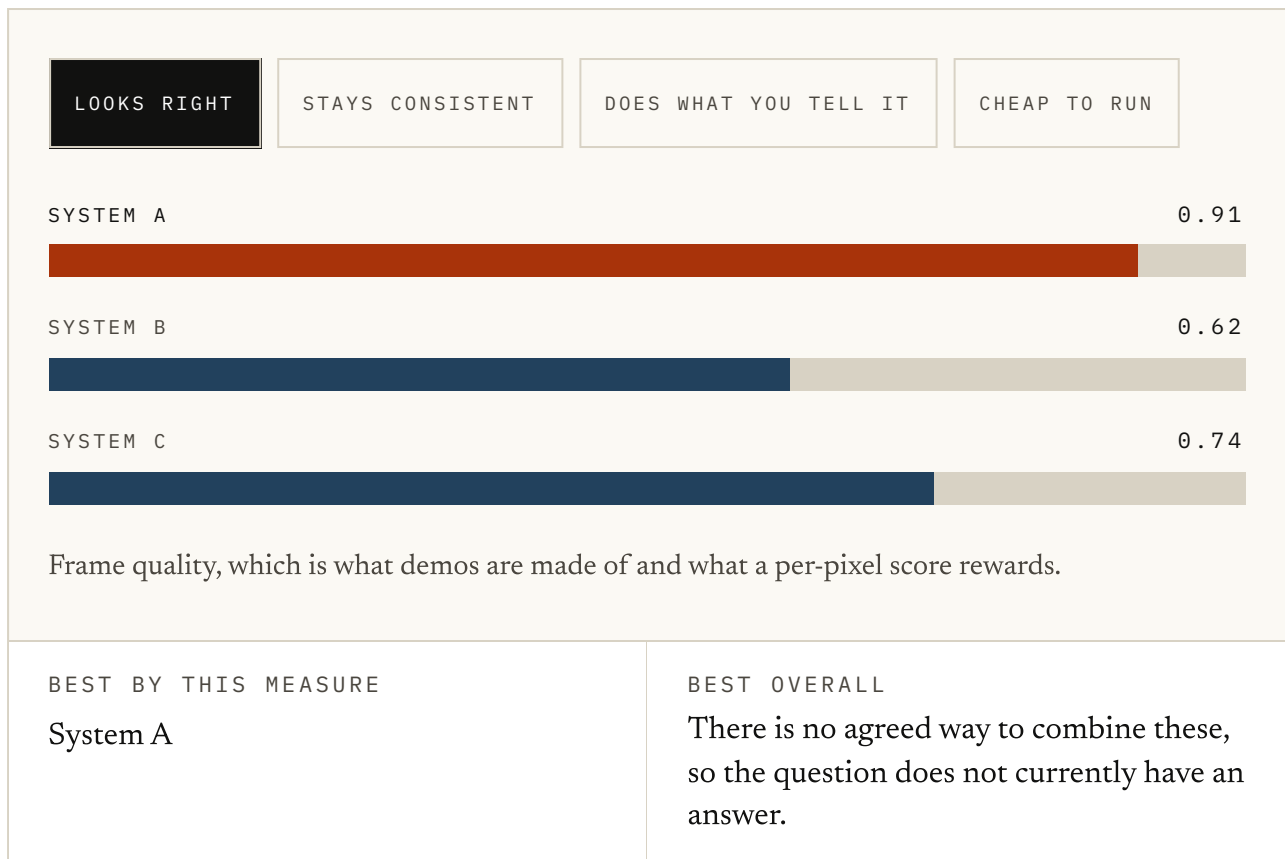


FIG. 9.2 The scores are invented and the systems are not real. The structure is: these measures genuinely disagree, and there is no accepted way to combine them into an ordering. Click through the four and watch the winner change.

Frame quality is measurable and is what demos are made of. Whether the world holds together over a thousand steps is what an agent needs, and no demo is long enough to show it. Whether the action you gave it is the action it took is the difference between a world model and a video, and it is almost never reported. Cost decides whether anything can afford to search inside the thing, and it is reported least of all.

Downstream task success looks like the honest answer, and it has its own problem: it scores the model and the policy together, so a good policy hides a bad model and the number tells you nothing about which you have.

None of this means progress is not happening. It means claims about progress are currently claims about whichever measure the claimant chose.

What to do with all this

Four questions, which between them handle most of what you will read.

What does it output? A picture, a structure, a compact state, an embedding. This decides what it can be used for and almost nothing else follows without it.

Can you roll it forward under actions nobody took? If not, it is not going to support a decision, however good it looks.

Over what horizon does it hold up, and who measured that? The number matters more than the architecture, and it is usually missing.

Which measure is the claim using, and what does that measure ignore? There is no overall score. Anyone implying there is has picked one and not told you.

That is the whole toolkit. It is unglamorous, and it will get you further through a paper than knowing any particular architecture.

Try it

1 Why is an average error over a long rollout close to useless as a report?

- a. Averages are always misleading
- b. The properties of a scene break in an order, and the ones that survive longest are the ones a per-frame score rewards
- c. The error is measured in the wrong units
- d. Rollouts are too short to average over

ANSWER b. The properties of a scene break in an order, and the ones that survive longest are the ones a per-frame score rewards. Colour and texture hold up almost indefinitely. Object identity goes first. So the average stays low long after the rollout has stopped being usable.

2 A rollout still looks completely fine in any single frame. What does that tell you?

- a. It is still usable
- b. Very little. Most of the pixels are surfaces and light, and nothing in a per-frame loss is watching whether objects stayed themselves
- c. The model has learned physics
- d. The horizon has not been reached

ANSWER b. Very little. Most of the pixels are surfaces and light, and nothing in a per-frame loss is watching whether objects stayed themselves. This is the specific way the subject is easy to fool yourself about, and it is why demos are weaker evidence than they feel.

3 Why is object permanence hard for these systems in particular?

- a. The models are too small
- b. Nothing is storing an object; consistency is a property of a good predictor, so it degrades gradually and without warning
- c. Occlusion is computationally expensive
- d. Training data lacks occluded objects

ANSWER b. Nothing is storing an object; consistency is a property of a good predictor, so it degrades gradually and without warning. There is no error message when the chair becomes a different chair. That is what it means for consistency to be learned rather than enforced.

4 You ask a model what would have happened if you had acted differently. What have you got?

- a. A fact about the world
- b. An answer that is true inside the model, with nothing in it indicating whether the model had data there
- c. A proof
- d. Nothing, models cannot answer that

ANSWER b. An answer that is true inside the model, with nothing in it indicating whether the model had data there. Where the model has data the two are close. Where it does not, they are not, and the model answers with equal confidence either way.

5 Short rollouts, replanning and starting from real states all work. What do they have in common?

- a. They make the model more accurate
- b. They are all ways of not needing the model to be right for very long
- c. They only apply to robotics
- d. They remove the need for a policy

ANSWER b. They are all ways of not needing the model to be right for very long. Read uncharitably, the whole toolkit is an admission. The horizon over which a model holds up is the number that decides what you can build.

6 Why does downstream task success not settle which model is better?

- a. Tasks are too easy
- b. It scores the model and the policy together, so a good policy hides a bad model
- c. It cannot be measured reliably
- d. It only works in simulation

ANSWER b. It scores the model and the policy together, so a good policy hides a bad model. It looks like the honest answer and it conflates the two things you were trying to tell apart.

7 Someone says their system is the best available. What is the first thing to ask?

- a. How large is it
- b. Which measure, and what does that measure ignore
- c. What hardware it runs on
- d. How long it trained for

ANSWER b. Which measure, and what does that measure ignore. There is no overall score. Frame quality, consistency over a long rollout, whether it obeys the action, and cost genuinely disagree, and anyone implying an ordering has picked one.

8 Across the whole course, which single question does the most work on an unfamiliar system?

- a. How many parameters does it have
- b. What does it output, and can you roll it forward under actions nobody took
- c. Is it open source
- d. Does it use a transformer

ANSWER b. What does it output, and can you roll it forward under actions nobody took. The first half decides what it can be used for. The second decides whether it can support a decision. Almost everything else follows, and neither depends on knowing the architecture.

FIG. 9.3 Eight questions on the honest state of it. If you have read the whole course, the last one should be easy and slightly annoying.

That is the course.

Nine chapters ago the phrase meant five different things and nobody said which. It still does. What has changed is that you can now tell which one is on the table, and what it would take for the claim to be true. And what to ask when somebody is not saying.

Thanks for reading. If something here is wrong, and some of it will be, the correction is genuinely welcome and will be credited.

Sources

How Far is Video Generation from World Model: A Physical Law Perspective

KANG ET AL., 2024 ↗

Physical laws matched inside the training distribution and not extrapolated outside it. The clearest measured statement of the gap this chapter is about.

<https://arxiv.org/abs/2411.02385>

VBench: Comprehensive Benchmark Suite for Video Generative Models

HUANG ET AL., 2023 ↗

A serious attempt at the measurement problem, and useful for seeing how many separate dimensions it takes before the ordering stops being obvious.

<https://arxiv.org/abs/2311.17982>

Benchmarking Model-Based Reinforcement Learning WANG ET AL., 2019 ↗

Where model-based methods win and lose, and the discovery that the planning horizon is the number that decides it.

<https://arxiv.org/abs/1907.02057>

When to Trust Your Model: Model-Based Policy Optimisation JANNER ET AL., 2019 ↗

Short rollouts as an answer to a model you cannot trust for long. The title is this chapter's question.

<https://arxiv.org/abs/1906.08253>

Emergent World Representations LI ET AL., 2022 ↗

The strongest available evidence that something structured forms inside a predictor, and a good example of what it takes to show it rather than assert it.

<https://arxiv.org/abs/2210.13382>

Genie 3 GOOGLE DEEPMIND, 2025 ↗

Reported coherence over minutes, from the lab that built it. Included as an example of exactly the kind of claim this chapter is asking you to read carefully.

<https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>

A Path Towards Autonomous Machine Intelligence LECUN, 2022 ↗

The most complete statement of what a world model would have to do to be worth the name, which doubles as a list of what is still missing.

<https://openreview.net/forum?id=BZ5a1r-kVsf>

FIG. 9.4 Ordered for someone building on this rather than for historical completeness.